Modeling and Forecasting VN-Index Volatility: A Robust Empirical Comparison of GARCH-Family, Neural, and Hybrid Approaches

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Abstract—This paper provides a validation-selected, proxy-robust, and risk-aware empirical benchmark for one-step-ahead daily VN-Index volatility forecasting using CafeF VNINDEX data. The target is next-day squared percentage log return, and the comparison covers GARCH(1,1), ARIMA-GARCH, a broad advanced GARCH-family search, EWMA/RiskMetrics, HAR-type baselines, LSTM and hybrid neural forecasts, calibration and forecast combinations, VaR backtesting, and OHLC proxy robustness. Stage 1 strengthens benchmark completeness: EWMA selected $\lambda = 0.90$ by validation QLIKE and obtains test QLIKE 1.225435, while HAR-Parkinson achieves test QLIKE 1.034521, making it competitive with the best advanced GARCH-family forecasts. The advanced search attempted 3,168 candidates and produced 2,592 successful fits; the primary advanced result remains the validation-selected AdvGARCH-BestQLIKE model with test QLIKE 1.032548, while test-best and category-best specifications are treated as exploratory. Statistical evidence remains conservative: raw QLIKE improvements favor advanced GARCH-family models, but no advanced GARCH improvement survives Holm correction at the 10% level and an approximate MCS retains many models. Risk diagnostics alter the interpretation of average loss: the 1% common-Normal VaR backtest for GARCH(1,1) has 16 violations out of 745 observations with Kupiec $p = 0.00631$, rejecting unconditional coverage despite acceptable 5% calibration. Proxy analysis shows that exact rankings are proxy-sensitive, but asymmetric GARCH-family variants remain robust top performers. Neural and hybrid forecasts may contain complementary information, especially after calibration, but they do not robustly dominate econometric benchmarks. Overall, GARCH-family and HAR-type models are more reliable under volatility-specific and risk-aware evaluation.

Index Terms—VN-Index, volatility forecasting, GARCH, EGARCH, LSTM, hybrid forecasting, QLIKE, VaR backtesting, range-based volatility.

I. Introduction

Volatility forecasting is an economic risk problem. For an equity index, a one-step-ahead variance forecast summarizes the expected scale of the next return shock and feeds directly into risk limits, margin setting, portfolio risk controls, Value-at-Risk (VaR), and tail-risk monitoring. A forecast that is accurate on average but too low during turbulent periods can be operationally dangerous, because risk systems are most useful precisely when volatility spikes.

The VN-Index provides a meaningful setting for this question. It is an emerging-market equity index with visible volatility clustering and sharp shocks, but without the routine availability of high-frequency realized-volatility data used in many developed-market studies. Daily open, high, low, close, and volume data therefore support a realistic empirical design: forecasts must be made from daily information, and the latent conditional variance must be evaluated through noisy observable proxies.

This paper is not a loose contest among unrelated models. The true daily volatility is unobserved, squared returns are noisy, and OHLC range-based proxies can change model rankings because they measure related but non-identical volatility objects. Model choice should therefore be validation-based, proxy-aware, and judged not only by average error but also by regime behavior, spike underprediction, and VaR coverage. The paper is framed as a proxy-robust and risk-aware empirical benchmark for daily VN-Index volatility forecasting.

The comparison covers historical and rolling variance baselines, GARCH(1,1), ARIMA-GARCH, advanced GARCH-family specifications, EWMA/RiskMetrics, HAR-type models, LSTM and ARIMA-GARCH-LSTM hybrids, tuned neural variants, validation-fitted calibration, and forecast combinations. The key empirical message is deliberately conservative: GARCH-family models remain the most reliable model class, while HAR-Parkinson becomes an important strong benchmark and neural/hybrid forecasts do not robustly dominate econometric alternatives.

The contributions are as follows:

1) A reproducible chronological benchmark for one-step-ahead VN-Index volatility forecasting using daily CafeF VNINDEX data and a percentage-squared return target.
2) A benchmark-complete comparison including GARCH(1,1), ARIMA-GARCH, advanced GARCH-family models, EWMA/RiskMetrics, HAR-type baselines, LSTM, hybrid forecasts, calibration, and forecast combinations.
3) A risk-aware evaluation protocol using QLIKE, Diebold-Mariano tests, Holm correction, an approximate model confidence set, regime and spike diagnostics, and VaR backtesting.
4) A proxy-robustness analysis using close-to-close, Parkinson, Garman-Klass, Rogers-Satchell, and Yang-Zhang proxies, showing that exact rankings

are proxy-sensitive but GARCH-family and asymmetric GARCH-family variants remain robust top performers.

II. Related Work

ARCH and GARCH models provide the classical foundation for conditional volatility forecasting. Engle introduced ARCH to model time-varying conditional variance, and Bollerslev generalized the recursion to include lagged conditional variance [1], [2]. The empirical appeal of GARCH(1,1) is not only parsimony. Large-scale forecast comparisons have shown that it can be difficult to beat in practical volatility forecasting applications [3].

Mean dynamics are often modeled before conditional variance. The Box-Jenkins ARIMA framework captures serial dependence in univariate time series [4]. In financial returns, the conditional mean is usually small relative to the conditional variance, but an ARMA or ARIMA mean model can still be paired with GARCH to test whether residual variance forecasts improve after filtering predictable mean components.

EWMA/RiskMetrics is a standard industry benchmark for volatility and VaR forecasting because it provides a transparent exponentially weighted recursion with very few tuning choices [5]. Its inclusion matters empirically: if a sophisticated model does not clearly improve on EWMA under volatility-specific loss and tail diagnostics, the additional complexity is hard to justify.

HAR-type models provide another important benchmark class. The heterogeneous autoregressive realized-volatility model captures persistence at daily, weekly, and monthly horizons through a simple linear structure [6]. Even when high-frequency realized volatility is unavailable, HAR-style regressions using squared-return or OHLC range proxies are useful because they test whether multi-horizon volatility persistence explains daily variance better than naive rolling volatility.

GARCH-family extensions address stylized features that a symmetric GARCH(1,1) recursion may miss. EGARCH models allow asymmetric volatility responses and avoid some positivity restrictions [7]. GJR-type models capture sign-dependent shock effects [8]. APARCH and FIGARCH specifications allow richer power transformations and long-memory volatility dynamics [9], [10]. These extensions motivate the broad specification search evaluated in this paper.

Neural sequence models offer a different forecasting premise. LSTM networks were designed to learn long-range sequential dependencies through gated recurrent states [11]. In volatility forecasting, LSTMs can use lagged returns, volume, rolling volatility, and econometric forecasts as supervised features. Hybrid econometric-neural designs are therefore useful empirical candidates, but their performance must be judged against strong volatility-specific baselines rather than against weak naive alternatives.

TABLE I
Chronological data splits and common evaluation window. The common-window comparison aligns all original forecasts on 726 target dates from 2023-02-07 to 2025-12-31.

Dataset	Rows	Origin date range	Target date range
Prepared full	3989	2010-01-04 to 2025-12-31	2010-01-05 to 2025-12-31
Model-ready	3968	2010-02-01 to 2025-12-30	2010-02-02 to 2025-12-31
Train	2471	2010-02-01 to 2019-12-30	2010-02-02 to 2019-12-31
Validation	750	2020-01-02 to 2022-12-29	2020-01-03 to 2022-12-30
Test	745	2023-01-03 to 2025-12-30	2023-01-04 to 2025-12-31
Common test	726	2023-02-06 to 2025-12-30	2023-02-07 to 2025-12-31

Forecast evaluation is central because latent volatility is not directly observed. Squared daily returns are available and simple, but they are noisy proxies for conditional variance. QLIKE is widely used because it remains attractive when forecast comparisons rely on imperfect volatility proxies and because it evaluates forecasts on the variance scale [12]. High-frequency realized volatility can provide richer targets when intraday data are available [13], but daily OHLC estimators are also useful. Parkinson, Garman-Klass, Rogers-Satchell, and Yang-Zhang estimators exploit high, low, open, and close information to obtain alternative volatility proxies [14]–[17].

Statistical forecast comparisons require care when many models are examined. Diebold-Mariano tests compare predictive accuracy under a specified loss difference process [18]. Holm correction controls family-wise error rates across multiple tests [19]. The model confidence set (MCS) framework provides a complementary way to identify a set of models not statistically eliminated under a loss criterion [20]. VaR backtesting adds a risk-management view through unconditional and conditional coverage tests [21], [22].

III. Data and Forecasting Target

A. Daily VN-Index Data

The empirical sample uses CafeF VNINDEX daily observations from 2010 through 2025. The cleaned data include open, high, low, close, matched trading volume, and matched trading value. Table I reports the chronological splits. The model-ready sample contains 3968 rows from 2010-02-01 to 2025-12-30 after lagged-return, rolling-volatility, and next-day-target requirements are imposed.

The experimental split is chronological. Training observations end in 2019, validation covers 2020–2022, and the test period begins in 2023. The original common-window comparison uses 726 aligned test observations with target dates from 2023-02-07 to 2025-12-31, so every original model is evaluated on identical forecast origins and labels.

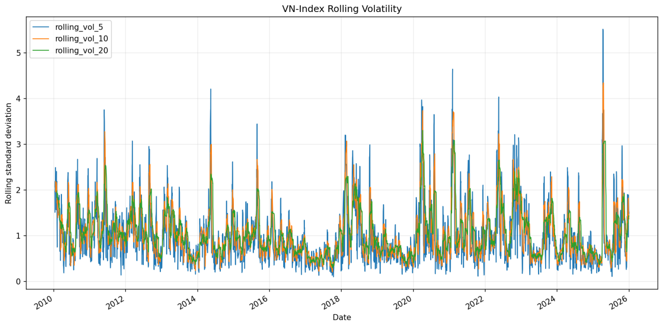

Fig. 1. Rolling VN-Index volatility changes materially across the sample, motivating conditional-variance models rather than constant-variance forecasts.

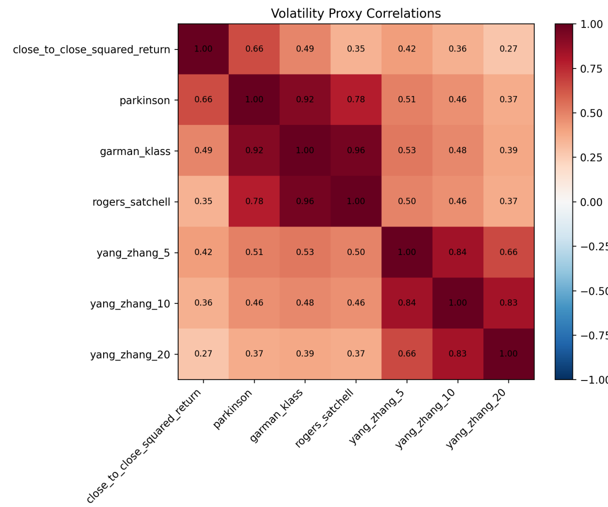

Fig. 2. Stage 1 proxy correlations show that squared returns and OHLC range-based proxies measure related but non-identical volatility objects, motivating proxy-robust rather than single-proxy claims.

B. Squared-Return Target

Let C_t denote the VN-Index close on trading day t. Percentage log return is

$$r_t = 100 \log \left(\frac{C_t}{C_{t-1}} \right), \tag{1}$$

and the daily squared-return variance proxy is

$$v_t = r_t^2. \tag{2}$$

The supervised one-step target is

$$y_t = v_{t+1} = r_{t+1}^2. \tag{3}$$

Thus a forecast issued at origin date t uses information available through t to predict the target date $t+1$.

The return r_t is measured in percentage log-return units, so v_t and y_t are percentage-squared variance proxies. Squared daily return is observable and convenient, but it is a noisy measure of latent volatility. A single daily return can be small even when intraday variation is high, and a single jump can dominate the proxy. This motivates QLIKE-based evaluation and robustness checks with OHLC range-based volatility proxies. Exact model rankings are proxy-sensitive, so the paper emphasizes class-level conclusions rather than a single universal winner.

C. OHLC Volatility Proxies

The proxy robustness analysis constructs seven volatility proxies from the cleaned OHLC sample: close-to-close squared return, Parkinson, Garman-Klass, Rogers-Satchell, and Yang-Zhang rolling proxies over 5, 10, and 20 days. These proxies are not treated as ground truth replacements for the squared-return target. Instead, they provide alternative noisy measurements of latent volatility and test whether model conclusions survive reasonable changes in the target proxy.

Stage 1 cleaned the proxy correlation and rank-stability artifacts. The correlations between close-to-close squared return and range-based proxies are moderate or low: 0.656 for Parkinson, 0.489 for Garman-Klass, 0.346 for Rogers-Satchell, 0.417 for Yang-Zhang 5-day, 0.359 for Yang-Zhang 10-day, and 0.274 for Yang-Zhang 20-day. These values explain why range-based robustness can change exact model orderings while still supporting class-level conclusions.

IV. Models

A. Simple Variance Baselines

The historical mean baseline predicts the training-sample average variance. Rolling baselines use 5-, 10-, and 20-day trailing return volatility, squared to produce variance forecasts. These simple baselines are important because volatility is persistent and recent realized variation can be competitive over short horizons.

B. EWMA/RiskMetrics

The EWMA baseline follows the RiskMetrics-style recursion

$$\sigma_{t+1}^2 = \lambda \sigma_t^2 + (1-\lambda) r_t^2, \tag{4}$$

where r_t^2 is the current squared percentage return. Stage 1 evaluates $\lambda \in \{0.90, 0.94, 0.97, 0.99\}$ and selects λ by validation QLIKE. EWMA is deliberately simple, but it is an important industry baseline for comparing volatility forecasts and Normal VaR mappings.

C. HAR-Type Baselines

HAR models approximate heterogeneous volatility persistence with daily, weekly, and monthly components:

$$y_{t+1} = \beta_0 + \beta_d RV_{d,t} + \beta_w RV_{w,t} + \beta_m RV_{m,t} + \varepsilon_{t+1}. \tag{5}$$

Here $RV_{d,t}$ is the most recent daily volatility proxy, while $RV_{w,t}$ and $RV_{m,t}$ are trailing weekly and monthly averages. Stage 1 implements HAR-SquaredReturn, Log-HAR-SquaredReturn, and HAR-Parkinson variants, with the final variant selected by validation QLIKE. HAR is a stronger benchmark than naive rolling volatility because it models multi-horizon persistence directly.

D. GARCH(1,1) and ARIMA-GARCH

GARCH(1,1) models conditional variance as

$$r_t = \mu + \epsilon_t, \qquad \epsilon_t = \sigma_t z_t, \tag{6}$$

$$\sigma_t^2 = \omega + \alpha\epsilon_{t-1}^2 + \beta\sigma_{t-1}^2. \tag{7}$$

The parameters are estimated on the training split. Validation and test forecasts use fixed estimated parameters while recursively updating the volatility state with returns observed before each forecast origin.

ARIMA-GARCH first models the conditional mean of returns using ARIMA and then fits GARCH(1,1) to the ARIMA residuals. The implemented ARIMA order selection considers $p, q \in \{0, 1, 2, 3\}$ with $d = 0$ using training AIC, selecting ARIMA(0,0,3) for the original benchmark. This design tests whether filtering mean dynamics adds volatility-forecasting value beyond direct GARCH variance modeling.

E. Advanced GARCH-Family Search

The advanced search evaluates direct ARCH-family and two-step ARIMA-GARCH candidates. It includes ARCH, GARCH, EGARCH, GJR-GARCH, APARCH, and FIGARCH-type specifications with Normal, Student-t, skewed-t, and GED innovation distributions where supported by the fitting library. Candidate selection uses validation QLIKE. Test results are reported only after a model or category has been selected on validation data.

The search attempts 3,168 GARCH-family candidates. It records 2,592 successful fits and 576 failures, all classified as numerical errors such as non-finite forecasts. The best overall validation-selected GARCH-family model is AdvGARCH-BestQLIKE, an ARIMA(3,0,3)-EGARCH(1,1,2)-Normal specification. Category selections also identify the best asymmetric, heavy-tail, and symmetric variants.

F. Neural, Hybrid, Calibration, and Combination Models

The base LSTM uses length-20 sequences of lagged market and volatility features, including returns, squared returns, absolute returns, rolling volatility, volume, and trading value. The ARIMA-GARCH-LSTM hybrid augments these inputs with econometric variance and residual-related features. Tuned neural variants include log-target training and QLIKE-style loss training; selection is by validation QLIKE within the relevant neural group.

Log-target models train on transformed variance targets and invert predictions back to the variance scale. This can stabilize training, but it can also produce low and smooth forecasts if scale is not recovered well. Calibration methods therefore include multiplicative QLIKE scaling, mean matching, nonnegative affine calibration, and isotonic calibration fitted on validation data only. Forecast combinations include equal-weight means, medians, two-model convex combinations, and stacking weights selected by validation QLIKE.

G. Risk Forecasts

Variance forecasts are mapped to VaR forecasts under a common Normal-quantile rule so that different variance models can be compared under a shared risk backtesting convention. The backtests report violation counts, expected violation counts, Kupiec unconditional coverage statistics, and Christoffersen independence and conditional coverage diagnostics.

V. Experimental Protocol

A. Chronological Evaluation and Leakage Controls

All model development follows the chronological train-validation-test design in Table I. Training data are used for fitting parameters, scalers, and imputers. Validation data are used for neural selection, EWMA decay selection, HAR variant selection, advanced GARCH-family selection, calibration fitting, refit protocol selection, and combination-weight selection. Test data are reserved for final evaluation.

The target, target date, date, and split labels are excluded from predictive features. The forecast origin t predicts $t+1$, rolling features are trailing-only, and LSTM validation and test sequences are built within each split so no sequence crosses a split boundary. Econometric parameters are estimated on training data unless an explicitly documented refit protocol is being evaluated. Stage 1 added leakage and formula tests covering target alignment, proxy formulas, and VaR helpers; both pytest and make stage1-test passed with 22 tests.

B. Metrics and Forecast Comparison

For aligned observations $\{y_t, \hat{y}_t\}_{t=1}^{n}$, the evaluation uses

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}, \tag{8}$$

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|, \tag{9}$$

$$\text{QLIKE} = \frac{1}{n}\sum_{t=1}^{n}\left(\log(\hat{y}_t) + \frac{y_t}{\hat{y}_t}\right). \tag{10}$$

Predicted variances are clipped at $\epsilon = 10^{-8}$ before QLIKE computation. QLIKE is central because severe underprediction is penalized through $y_t/\hat{y}_t$. MAE is also reported, but it can be misleading when smooth low forecasts perform well on calm days yet fail during volatility spikes.

Diebold-Mariano tests use the loss-difference convention

$$d_t = L_{m,t} - L_{b,t}, \tag{11}$$

where $L_{m,t}$ is the compared model loss and $L_{b,t}$ is the benchmark loss. A positive mean difference means the compared model is worse than the benchmark. Multiple testing is summarized with Holm correction, and a documented moving-block bootstrap approximation is used for the model confidence set analysis.

C. Selection and Diagnostic Protocol

The primary selection rule is validation QLIKE. EWMA selects λ, HAR selects the volatility-proxy variant, and the broad advanced GARCH search selects the primary AdvGARCH-BestQLIKE model on validation data. Category-best and test-best combinations are useful diagnostics, but test-best combinations are exploratory and must not be treated as primary selected winners.

Stage 1 adds regime metrics, spike diagnostics, extended common-Normal VaR backtests at 5%, 2.5%, and 1%, cleaned proxy robustness artifacts, and an existing-prediction time-split diagnostic. The time-split diagnostic is not true rolling-origin retraining. White Reality Check and Hansen SPA tests are not implemented, and distribution-specific VaR is not computed because fitted shape and skew parameters are not retained in the current prediction artifacts.

VI. Results

A. Original Benchmark Results

Table II reports the original 726-observation common-window comparison. GARCH(1,1) has the lowest QLIKE, 1.052888, and the lowest RMSE, 3.648839. ARIMA-GARCH is very close, with QLIKE 1.055679, and the QLIKE Diebold-Mariano test against GARCH gives $p = 0.399560$. This is consistent with limited standalone value from ARIMA mean filtering for next-day variance forecasting.

The LSTM and ARIMA-GARCH-LSTM forecasts do not beat GARCH(1,1) by QLIKE. The log-target variants tell a different metric story: Hybrid-LogTarget-Small has the lowest MAE, 1.108165, but QLIKE rises to 2.232917 and mean predicted variance falls to 0.333670 against mean actual variance 1.211912. MAE alone is not a reliable volatility-forecasting criterion.

B. Benchmark Completeness: EWMA and HAR

Table III adds the Stage 1 benchmark-completeness results. The selected EWMA/RiskMetrics baseline uses $\lambda = 0.90$, with validation QLIKE 1.687209 and test QLIKE 1.225435. This is a useful industry benchmark, but it does not beat the best GARCH-family or HAR-type forecasts.

HAR is more important. HAR-Parkinson is selected by validation QLIKE, with validation QLIKE 1.558162 and test QLIKE 1.034521. HAR-Parkinson achieves test QLIKE 1.034521, making it competitive with the best advanced GARCH-family forecasts. This prevents the benchmark from being only a GARCH-versus-neural comparison and shows that range-based persistence contains substantial information for daily VN-Index volatility.

C. Advanced GARCH Search Audit

The advanced GARCH-family search attempts 3,168 candidates, records 2,592 successful fits, and has 576 failed fits. The failures are numerical rather than evidence against the model class. They do, however, reinforce the need for conservative selection and transparent audit reporting.

Table V summarizes the selected advanced specifications. The validation-selected overall model, AdvGARCH-BestQLIKE, obtains validation QLIKE 1.611304 and full-window test QLIKE 1.032548. AdvGARCH-BestAsymmetric obtains a lower standalone test QLIKE, 1.029992, but that comparison is exploratory because it is a category-selected specification rather than the single primary validation-selected winner. The primary advanced GARCH result remains validation-selected; test-best specifications are exploratory.

Raw QLIKE Diebold-Mariano tests show several advanced GARCH-family improvements against GARCH(1,1). AdvGARCH-BestAsymmetric has raw $p = 0.001534$, and AdvGARCH-BestQLIKE has raw $p = 0.018111$. Table VI shows why the interpretation remains cautious: no advanced GARCH improvement survives Holm correction at the 10% level, and the approximate MCS retains 68 of 73 models at both 90% and 95%.

D. Regime and Spike Diagnostics

Average QLIKE hides important risk behavior. Table VII shows that major models systematically underpredict high-volatility days. At the 90th-percentile spike threshold, GARCH(1,1), AdvGARCH-BestQLIKE, HAR-Parkinson, and EWMA all underpredict every spike day, and severe underprediction rates range from 86.7% to 88.9% for these core benchmarks. Raw log-target neural variants are worse: LSTM-LogTarget and Hybrid-LogTarget-Small have 100% severe spike underprediction in the reported Stage 1 diagnostic and spike-day pred/actual ratios near 0.06.

Spike underprediction is the main practical failure mode in daily volatility forecasting. The regime results do not overturn the average-loss conclusion that GARCH-family and HAR-type forecasts are strong. Instead, they show that even strong average models remain too smooth during the most important risk episodes.

E. Extended VaR Backtesting

Table VIII extends the risk evaluation to 5%, 2.5%, and 1% common-Normal VaR. GARCH(1,1) looks acceptable at 5%: it has 35 violations out of 745 observations against 37.25 expected and Kupiec $p = 0.7025$. The stricter tail is different. At the 1% tail level, GARCH(1,1) no longer

TABLE II
Original common-window test comparison. GARCH(1,1) is numerically best by QLIKE and RMSE, while log-target neural variants obtain low MAE at the cost of much worse QLIKE and lower mean predicted variance.

Rank	Model	N	RMSE	MAE	QLIKE	Mean predicted variance
1	GARCH(1,1)	726	3.648839	1.393909	1.052888	1.241320
2	ARIMA-GARCH	726	3.671816	1.400256	1.055679	1.236080
3	LSTM	726	3.769716	1.639734	1.161001	1.568589
4	ARIMA-GARCH-LSTM	726	3.763253	1.720113	1.168609	1.701791
5	LSTM-QLIKE	726	3.782485	1.714073	1.185876	1.666980
6	HistoricalMean	726	3.799706	1.449153	1.192259	1.198773
7	Hybrid-QLIKE	726	3.807296	1.862578	1.230508	1.877606
8	RollingVol-20	726	3.901051	1.462196	1.280635	1.208200
9	RollingVol-10	726	3.974129	1.478891	1.309964	1.218541
10	RollingVol-5	726	4.011383	1.492394	1.928009	1.201815
11	Hybrid-LogTarget	726	3.867813	1.109300	2.086360	0.339489
12	Hybrid-LogTarget-Small	726	3.868068	1.108165	2.232917	0.333670
13	LSTM-LogTarget-Small	726	3.888176	1.123546	2.281499	0.342765
14	LSTM-LogTarget	726	3.866882	1.108626	2.301502	0.309520

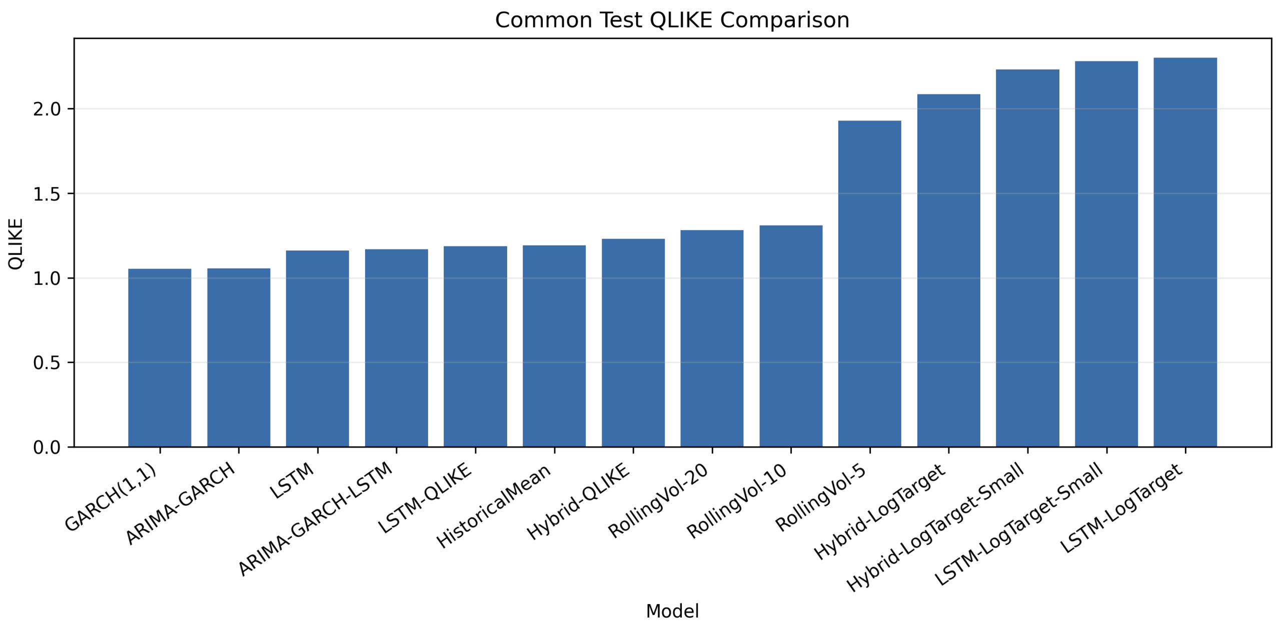

Fig. 3. Common-window QLIKE ranks the original econometric benchmarks ahead of the neural and hybrid variants, while raw log-target models are strongly penalized for variance underprediction.

TABLE III
Stage 1 benchmark-completeness additions. EWMA and HAR are selected only by validation QLIKE, then evaluated on the untouched test split. HAR-Parkinson is a strong benchmark rather than a cosmetic baseline.

Model	Validation selection rule	Val. QLIKE	Test QLIKE	N_{test}	Interpretation
EWMA($\lambda = 0.90$)	Best among $\lambda \in \{0.90, 0.94, 0.97, 0.99\}$	1.687209	1.225435	745	Industry-style RiskMetrics baseline; improves benchmark completeness but trails stronger volatility models
HAR-Parkinson	Best HAR variant by validation QLIKE	1.558162	1.034521	745	Competitive with advanced GARCH-family forecasts on the main squared-return target
HAR-SquaredReturn	Included HAR robustness variant	1.647660	1.047975	745	Stronger than naive rolling volatility, but not validation-selected
AdvGARCH-BestQLIKE	Best advanced GARCH-family candidate by validation QLIKE	1.611304	1.032548	745	Primary advanced GARCH-family selected model

passes unconditional coverage under the common Normal VaR mapping. It has 16 violations out of 745 observations against 7.45 expected, with Kupiec $p = 0.00631$.

This result should not be read as a rejection of GARCH-family volatility forecasts overall. It shows that even strong variance models need tail-specific validation. The common

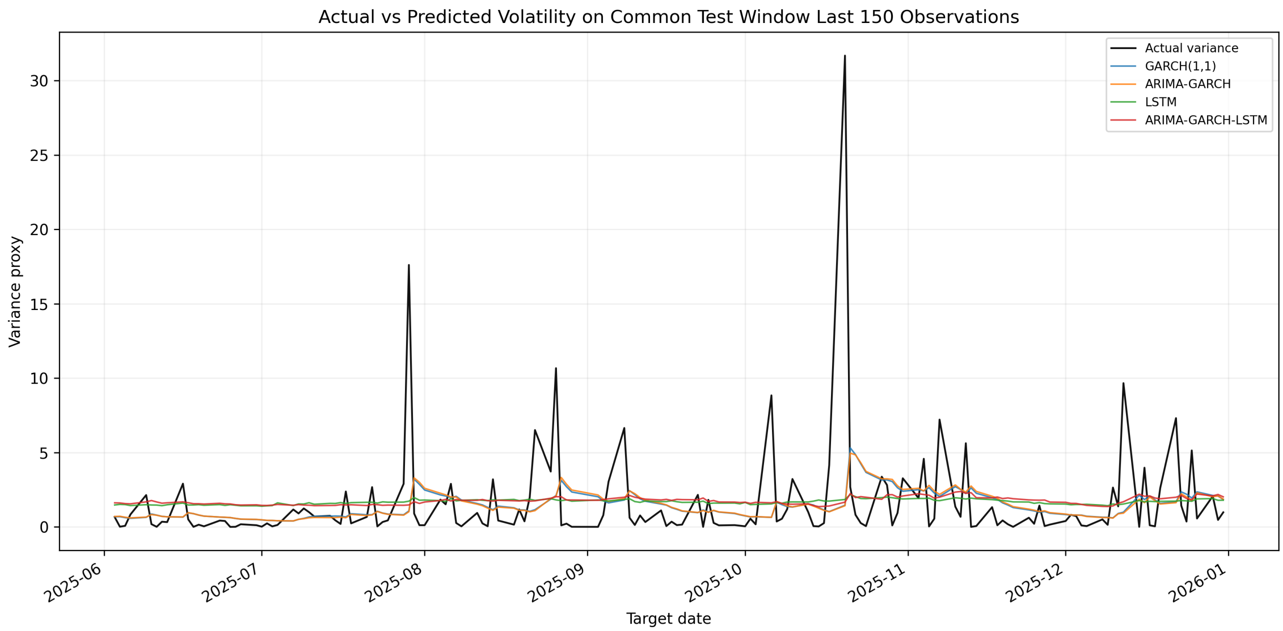

Fig. 4. Actual squared-return variance is spiky, while forecasts are smoother. Low smooth forecasts can reduce MAE on calm days but receive large QLIKE penalties when spikes are missed.

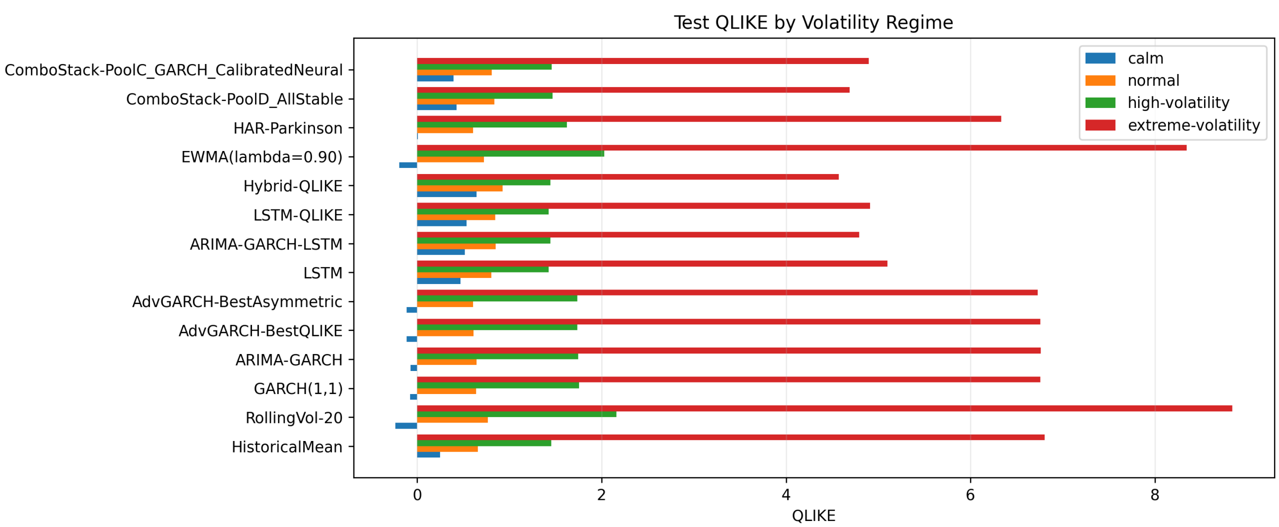

Fig. 5. Stage 1 regime QLIKE diagnostics show that model rankings differ between calm and extreme-volatility days; average QLIKE alone is not sufficient for risk interpretation.

TABLE IV
Advanced GARCH search audit. The broad search improves coverage of plausible econometric specifications, but the primary result remains validation-selected.

Audit item	Count
Attempted candidates	3,168
Successful fits	2,592
Failed fits	576
Convergence failures	0
Non-finite forecast failures	144

Normal mapping standardizes comparison across models, but it may be too restrictive for heavy-tailed or skewed GARCH-family specifications whose fitted distributional parameters are not retained in the current prediction artifacts.

F. Proxy Robustness

Proxy robustness changes exact rankings. The cleaned Stage 1 rank-stability rows show AdvGARCH-BestAsymmetric with the best average proxy rank, 3.57, top-5 placement under 6 of 7 proxies, and top-3 placement

TABLE V
Validation-selected advanced GARCH-family specifications. Test QLIKE values use the full advanced test window reported in the selected-model tables; statistical tests later align losses on their common window.

Model	Specification	Selection role	Val. QLIKE	Test QLIKE	Paper use
AdvGARCH-BestQLIKE	ARIMA(3,0,3)-EGARCH(1,1,2)-Normal	Overall validation QLIKE	1.611304	1.032548	Primary advanced GARCH-family selected model
AdvGARCH-BestAsymmetric	Zero-mean EGARCH(1,1,1)-Normal	Best asymmetric validation QLIKE	1.612272	1.029992	Category-selected model; best standalone test QLIKE is exploratory
AdvGARCH-BestHeavyTail	ARIMA(3,0,3)-EGARCH(1,1,1)-GED	Best heavy-tail validation QLIKE	1.611777	1.031536	Category-selected heavy-tail comparison
AdvGARCH-BestSymmetric	Constant-mean GARCH(1,3)-Normal	Best symmetric validation QLIKE	1.628556	1.063460	Symmetric robustness comparison

TABLE VI
Statistical comparison summary. Raw QLIKE differences sometimes favor advanced GARCH-family models, but Holm correction and the broad MCS prevent a decisive single-model superiority claim.

Comparison	Evidence	Result	Interpretation
Original QLIKE DM vs. GARCH(1,1)	ARIMA-GARCH $p = 0.399560$; rolling baselines, raw log-target variants, and Hybrid-QLIKE are worse at 10% or stronger	No significant ARIMA-GARCH difference	Mean dynamics add limited standalone value in the original benchmark set
Original squared-error DM vs. GARCH(1,1)	No model differs from GARCH at the 10% level	No robust MSE separation	RMSE rankings should be read descriptively
Original absolute-error DM vs. GARCH(1,1)	Log-target variants beat GARCH by MAE	MAE favors smoother low forecasts	Not sufficient for volatility-risk superiority because QLIKE and risk tests show underprediction
Advanced raw QLIKE DM vs. GARCH(1,1)	AdvGARCH-BestAsymmetric $p = 0.001534$; AdvGARCH-BestQLIKE $p = 0.018111$	Observed improvements before multiplicity correction	Evidence favors advanced GARCH-family forecasts, with caveat
Advanced Holm correction	No QLIKE improvement over GARCH(1,1) survives at 10%	No Holm-significant superiority	Do not claim a statistically decisive winner
Approximate MCS	Moving-block bootstrap approximation retains 68/73 models at 90% and 95%	Broad confidence set	Eliminates weak rolling/log-target models but retains many viable alternatives

TABLE VII
Stage 1 regime, spike, and 1% VaR diagnostics. Average QLIKE hides a common practical weakness: most forecasts remain too smooth on spike days, and tail coverage can fail even when average volatility loss is competitive.

Model	Extreme-regime QLIKE	Severe spike underpred.	Spike pred./actual	1% VaR violations	Interpretation
GARCH(1,1)	6.758	86.7%	0.273	16/745 ($p = 0.00631$)	Strong average benchmark, but strict tail coverage fails
AdvGARCH-BestQLIKE	6.756	88.9%	0.251	15/745 ($p = 0.01453$)	Primary advanced model; spike timing remains imperfect
HAR-Parkinson	6.335	88.9%	0.245	16/745 ($p = 0.00631$)	Competitive average QLIKE with similar tail undercoverage
EWMA($\lambda = 0.90$)	8.345	88.9%	0.266	18/745 ($p = 0.00101$)	Complete industry baseline, but weaker under extreme volatility
LSTM-LogTarget	20.256	100.0%	0.056	56/726 ($p = 3.81\times10^{-31}$)	Low smooth forecasts severely understate spikes and tails
Hybrid-LogTarget-Small	19.929	100.0%	0.060	48/726 ($p = 5.02\times10^{-24}$)	Low MAE does not translate into risk reliability

under 5 of 7 proxies. AdvGARCH-BestQLIKE remains robust, with average rank 5.14 and top-5 placement under 5 of 7 proxies. GARCH(1,1) remains competitive but less stable, with average rank 7.71 and top-5 placement under 2 of 7 proxies.

The stable conclusion is class-level: asymmetric GARCH-family models are robust top performers, but no single model wins under every proxy. HAR-Parkinson is highly competitive on the main squared-return target, yet it is not as stable across the generated OHLC proxy ranks. This reinforces the distinction between a strong benchmark and a universal proxy winner.

G. Neural and Hybrid Diagnostics

Neural and hybrid forecasts should not be dismissed. Calibration and pairwise combinations suggest that nonlinear or hybrid signals can contain complementary information. The best observed test combination, ComboPair-BestGARCHFamily_HybridQLIKE, obtains test QLIKE 1.026265 with weights 0.73 on AdvGARCH-BestARIMA and 0.27 on Hybrid-QLIKE. This is useful diagnostic evidence, but it is not a primary selected result because it is test-best.

The main neural limitation is reliability under volatility-specific and risk-aware evaluation. Raw log-target neu-

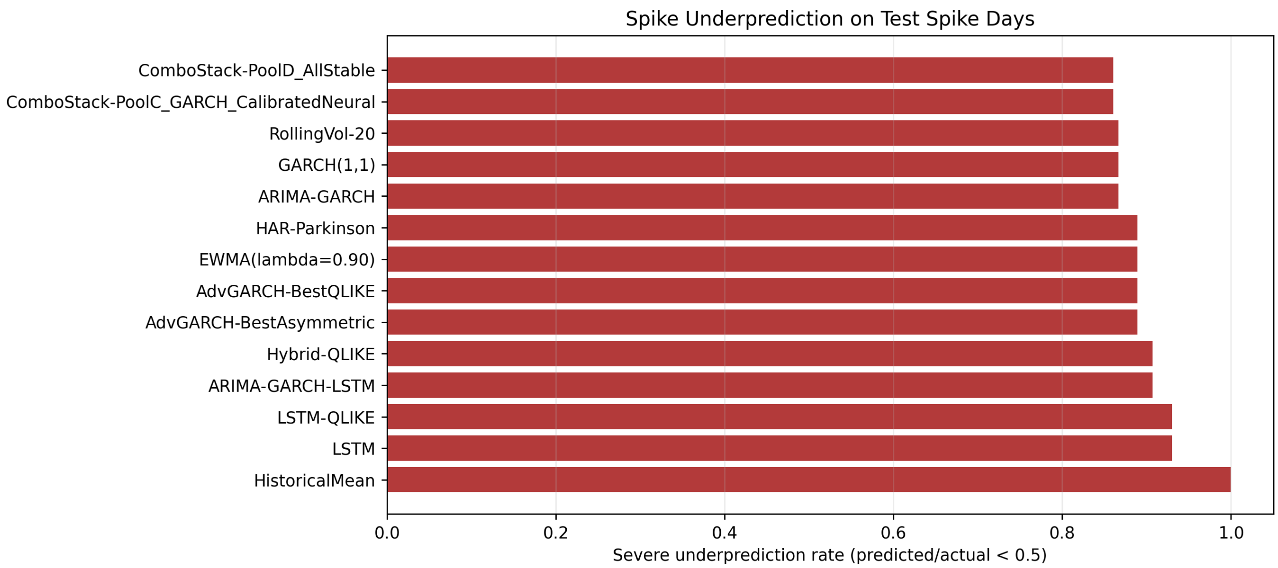

Fig. 6. Spike diagnostics identify underprediction as a common failure mode, with raw log-target neural variants showing the most severe spike-day scale errors.

TABLE VIII
Extended common-Normal VaR backtesting. GARCH(1,1) looks well calibrated at 5%, but the stricter 1% tail rejects unconditional coverage, showing why risk diagnostics must go beyond average QLIKE.

Model	VaR	N	Viol.	Exp.	Kupiec p	Indep. p	CC p	Interpretation
GARCH(1,1)	5%	745	35	37.25	0.7025	0.5709	0.7917	No 5% rejection
GARCH(1,1)	2.5%	745	22	18.63	0.4411	0.6778	0.6818	No 5% rejection
GARCH(1,1)	1%	745	16	7.45	0.00631	0.4017	0.0169	Rejects unconditional coverage
HAR-Parkinson	5%	745	33	37.25	0.4666	0.2386	0.3831	No 5% rejection
HAR-Parkinson	2.5%	745	17	18.63	0.6988	0.3726	0.6235	No 5% rejection
HAR-Parkinson	1%	745	16	7.45	0.00631	0.4017	0.0169	Rejects unconditional coverage
EWMA($\lambda = 0.90$)	5%	745	38	37.25	0.9000	0.9645	0.9912	No 5% rejection
EWMA($\lambda = 0.90$)	2.5%	745	27	18.63	0.0652	0.9832	0.1827	Borderline at 5%
EWMA($\lambda = 0.90$)	1%	745	18	7.45	0.00101	0.3448	0.00288	Rejects unconditional coverage
LSTM-LogTarget	5%	726	87	36.30	1.54×10^{-13}	0.0269	1.26×10^{-13}	Severe undercoverage
LSTM-LogTarget	1%	726	56	7.26	3.81×10^{-31}	0.0732	1.12×10^{-30}	Severe undercoverage

TABLE IX
Stage 1 volatility-proxy robustness rank summary. Exact rankings change across close-to-close and range-based proxies, but asymmetric GARCH-family specifications remain robust top performers.

Model	Avg. QLIKE rank	Best	Worst	Top-5 proxies	Interpretation
AdvGARCH-BestAsymmetric	3.57	1	14	6/7	Most stable top model; top-3 under 5/7 proxies
AdvGARCH-BestHeavyTail	4.71	2	15	5/7	Robust top model, but not a top-1 proxy winner
AdvGARCH-BestQLIKE	5.14	3	11	5/7	Primary validation-selected GARCH-family model remains robust
GARCH(1,1)	7.71	2	16	2/7	Strong benchmark, not consistently top-tier under range proxies
EWMA($\lambda = 0.90$)	13.14	1	58	3/7	Useful industry baseline, but proxy stability is weaker
HAR-Parkinson	15.14	7	22	0/7	Strong main-target benchmark, less stable across generated proxies
ComboPair-BestGARCHFamily_HybridQLIKE	16.71	1	20	1/7	Best close-to-close result, but proxy-sensitive

ral forecasts have average test QLIKE 2.225569 and pred/actual ratio 0.273; calibration improves the best

TABLE X

Calibration and forecast-combination summary. Calibration corrects much of the log-target scale bias, while validation-selected combinations do not establish robust hybrid superiority.

Result block	Selected or diagnostic item	QLIKE	Pred./actual	Interpretation
Raw log-target neural mean	Raw log-target group	2.225569	0.273	Low MAE comes with severe scale underprediction
Calibrated log-target best	Best validation-calibrated log-target summary	1.146217	1.496	Calibration materially improves QLIKE but remains above best GARCH-family QLIKE
Best validation combination	ComboStack-PoolD_AllStable	1.113530	1.398	Primary combination result; weights mostly calibrated log-target isotonic models
Best observed test combination	BestGARCHFamily + HybridQLIKE pair	1.026265	1.098	Exploratory test-best result; weights 0.73 on AdvGARCH-BestARIMA and 0.27 on HybridQLIKE
Best validation-selected GARCH-family	AdvGARCH-BestQLIKE	1.032548	0.936	Main advanced GARCH-family selected benchmark

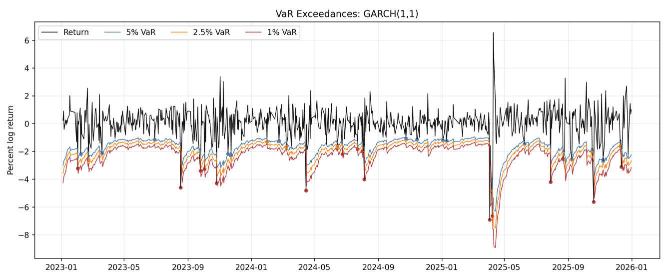

Fig. 7. GARCH(1,1) common-Normal VaR exceedances are acceptable at 5%, but stricter 1% coverage is too low in the Stage 1 extended backtest.

calibrated log-target QLIKE to 1.146217 and raises the pred/actual ratio to 1.496. Calibration improves neural scale bias, but spike timing and tail-risk reliability remain unresolved.

VII. Discussion

GARCH-family models remain the most reliable model class. Their advantage is structural: they directly encode volatility persistence, and the fitted GARCH persistence values near 0.97 indicate that this structure is empirically important for the VN-Index. Advanced asymmetric and EGARCH-type variants improve raw QLIKE and are robust across proxies, but the formal evidence is not strong enough to declare one specification universally dominant.

HAR-Parkinson materially strengthens the benchmark. Its test QLIKE of 1.034521 is close to the primary advanced GARCH-family result, and it shows that daily OHLC range information can capture useful multi-horizon volatility persistence. This makes the empirical comparison more credible because GARCH-family forecasts are evaluated against a strong HAR-type alternative, not only against rolling volatility.

The neural and hybrid evidence is mixed rather than negative. Daily sample size is modest for deep sequence models, and log-target training can reward smooth forecasts that reduce MAE while missing economically important spikes. Calibration corrects much of the scale bias, and hybrid combinations suggest complementary signals, but primary validation-selected neural or hybrid models do not robustly dominate econometric benchmarks.

Proxy robustness is essential because volatility is latent. Squared returns and OHLC range estimators measure related but non-identical objects. The empirical conclusion is therefore safer at the class level than at the individual model level: GARCH-family forecasts, especially asymmetric variants, are robust top performers, while exact ranks vary across proxies.

For applied VN-Index risk forecasting, model choice should be based on volatility-specific loss, spike behavior, and VaR coverage rather than MAE alone. Average QLIKE is necessary but not sufficient for risk use; spike underprediction and 1% VaR coverage determine whether a forecast is operationally credible.

VIII. Limitations

Several limitations qualify the findings without weakening the benchmark's transparency. The study uses daily data only and does not have high-frequency realized volatility. Squared returns and OHLC range estimators are imperfect proxies for latent volatility, so proxy robustness is informative but not a substitute for true realized-variance targets.

The advanced GARCH-family search is broad enough to create validation-overfitting risk, and 576 candidate fits fail numerically. The approximate MCS retains many models, so it should be read as conservative evidence against a decisive single winner rather than as a precise model ranking. True rolling-origin retraining was not run in Stage 1; only an existing-prediction time-split diagnostic was generated.

White Reality Check and Hansen SPA tests were not implemented. Distribution-specific VaR was not computed because fitted distribution shape and skew parameters are not retained in the current prediction artifacts. The common Normal VaR mapping standardizes comparison, but it may be too restrictive for heavy-tail models. Finally, the neural models are trained on a modest daily sample, so their failure to dominate should be interpreted as empirical fragility in this setting rather than a general claim against neural volatility forecasting.

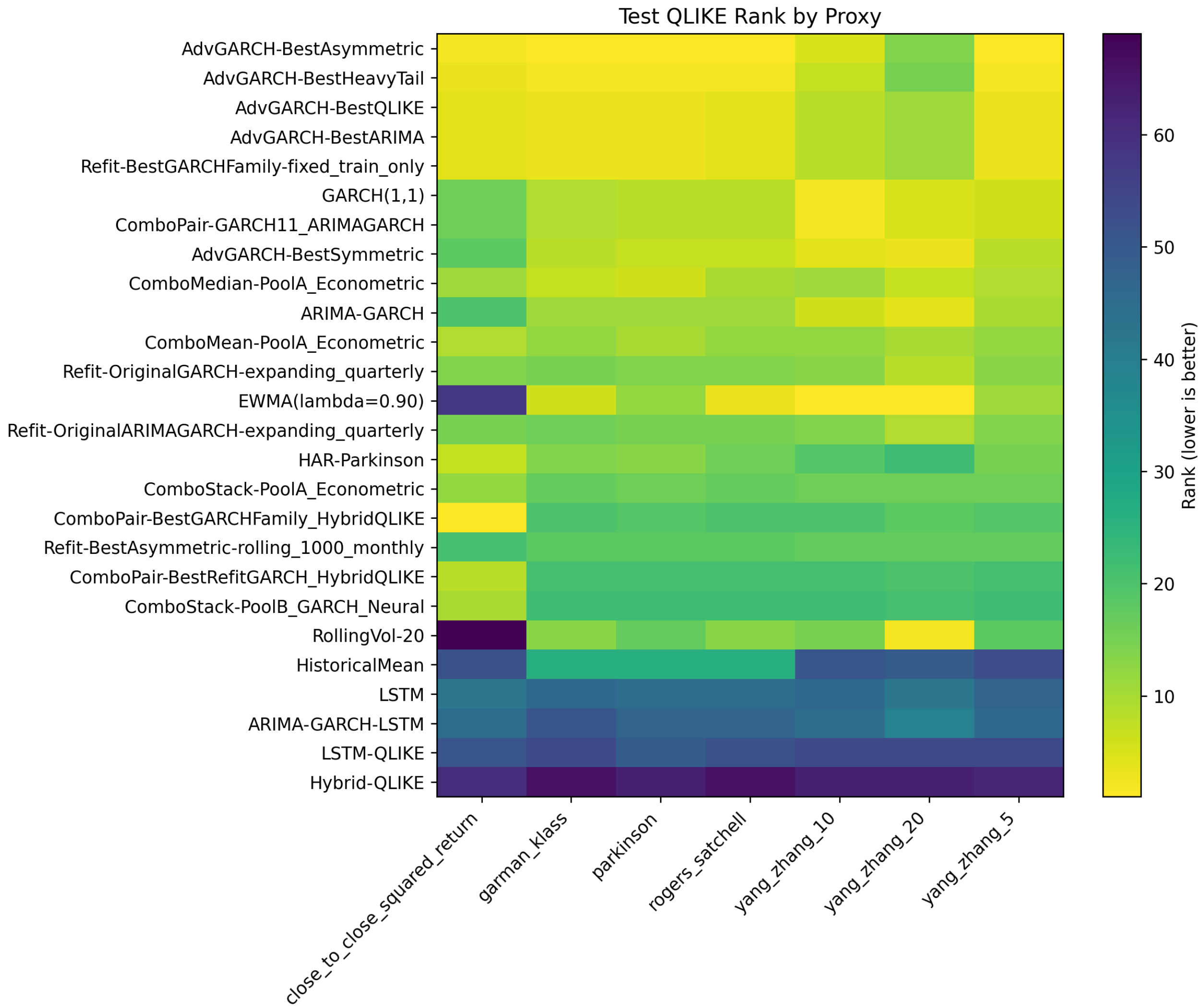

Fig. 8. Stage 1 proxy rank heatmap. Range-based proxies change exact ordering, but asymmetric GARCH-family specifications remain consistently near the top across the generated stability rows.

IX. Conclusion

This paper provides a validation-selected, proxy-robust, and risk-aware empirical benchmark for one-step-ahead daily VN-Index volatility forecasting. The evidence supports a conservative conclusion: GARCH-family models remain the most reliable class for daily VN-Index volatility forecasting, but the benchmark should be interpreted through validation selection, proxy robustness, and risk diagnostics rather than a single test-best ranking.

Stage 1 strengthens the empirical comparison. EWMA/RiskMetrics adds an industry baseline, while HAR-Parkinson becomes a serious competitor with test QLIKE 1.034521. The broad advanced GARCH-family search confirms that asymmetric and EGARCH-type specifications are robust top performers across OHLC proxies, although Holm correction and the approximate MCS prevent a decisive claim that one model dominates all others.

Neural and hybrid forecasts may contain complementary information, especially after calibration and in selected combinations, but they do not robustly dominate econometric benchmarks. Spike underprediction and 1% VaR undercoverage show why average QLIKE is not enough for risk use. Future work should add high-frequency realized volatility, true rolling-origin retraining, White Reality Check or Hansen SPA inference, distribution-specific VaR and expected shortfall, and stronger hybrid residual-correction models.

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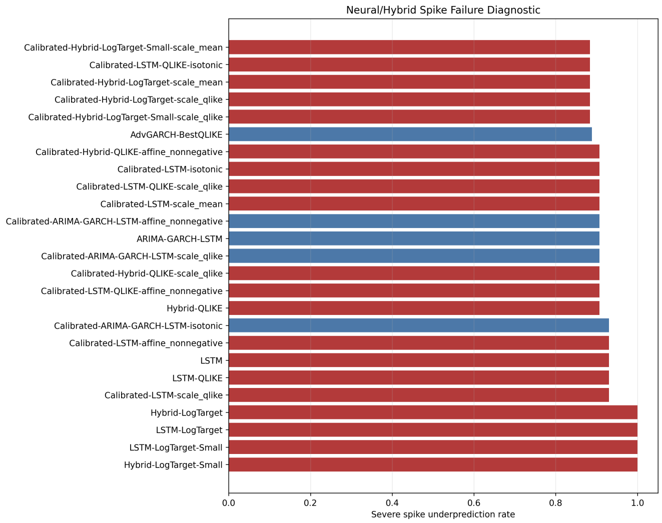

Fig. 9. Stage 1 neural spike diagnostic. Log-target neural forecasts can look attractive under MAE but remain too smooth during volatility spikes.

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